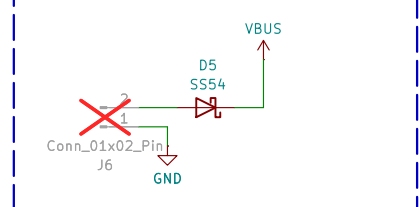
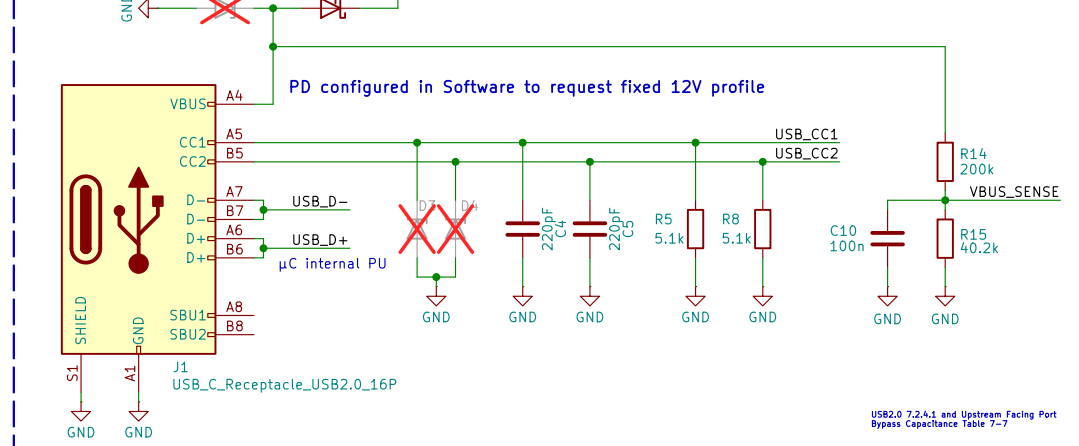


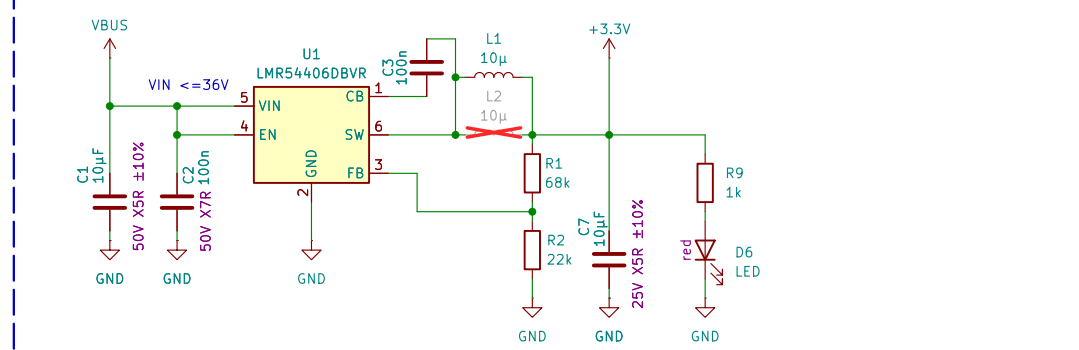
Board Power Supply (+12V)



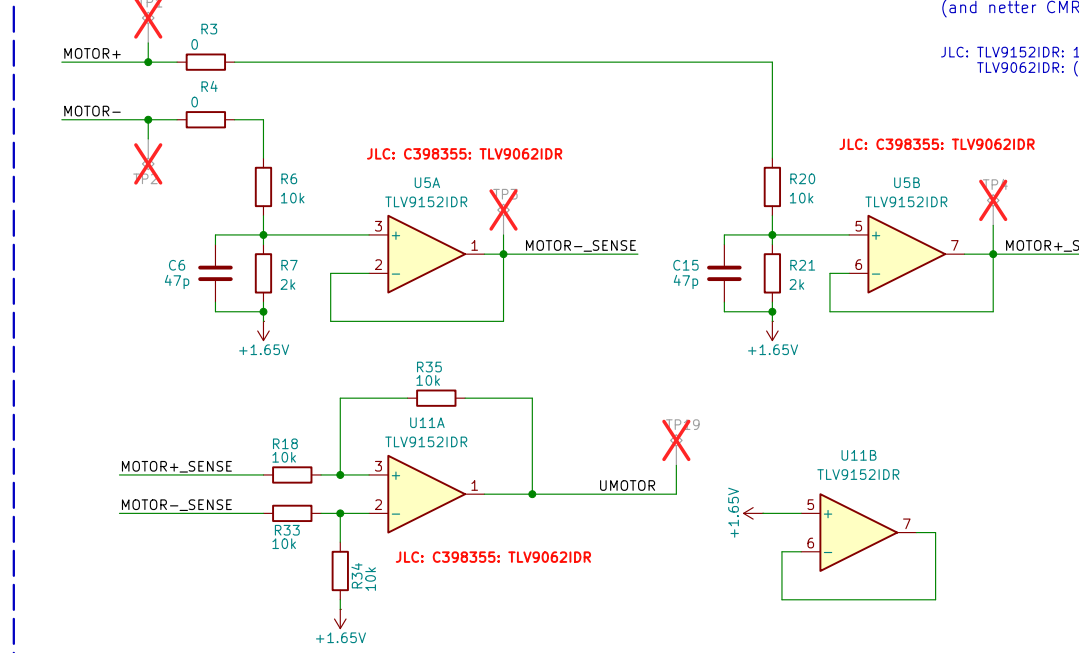
USB Data onboard $\mu C \rightarrow PC$. USBC Power OPTIONAL
NOT used for first experiments
only if we want to power the Board with 9/12V via USBC-PD later
(not sure if usbcd negotiation has trouble with D2/D5)



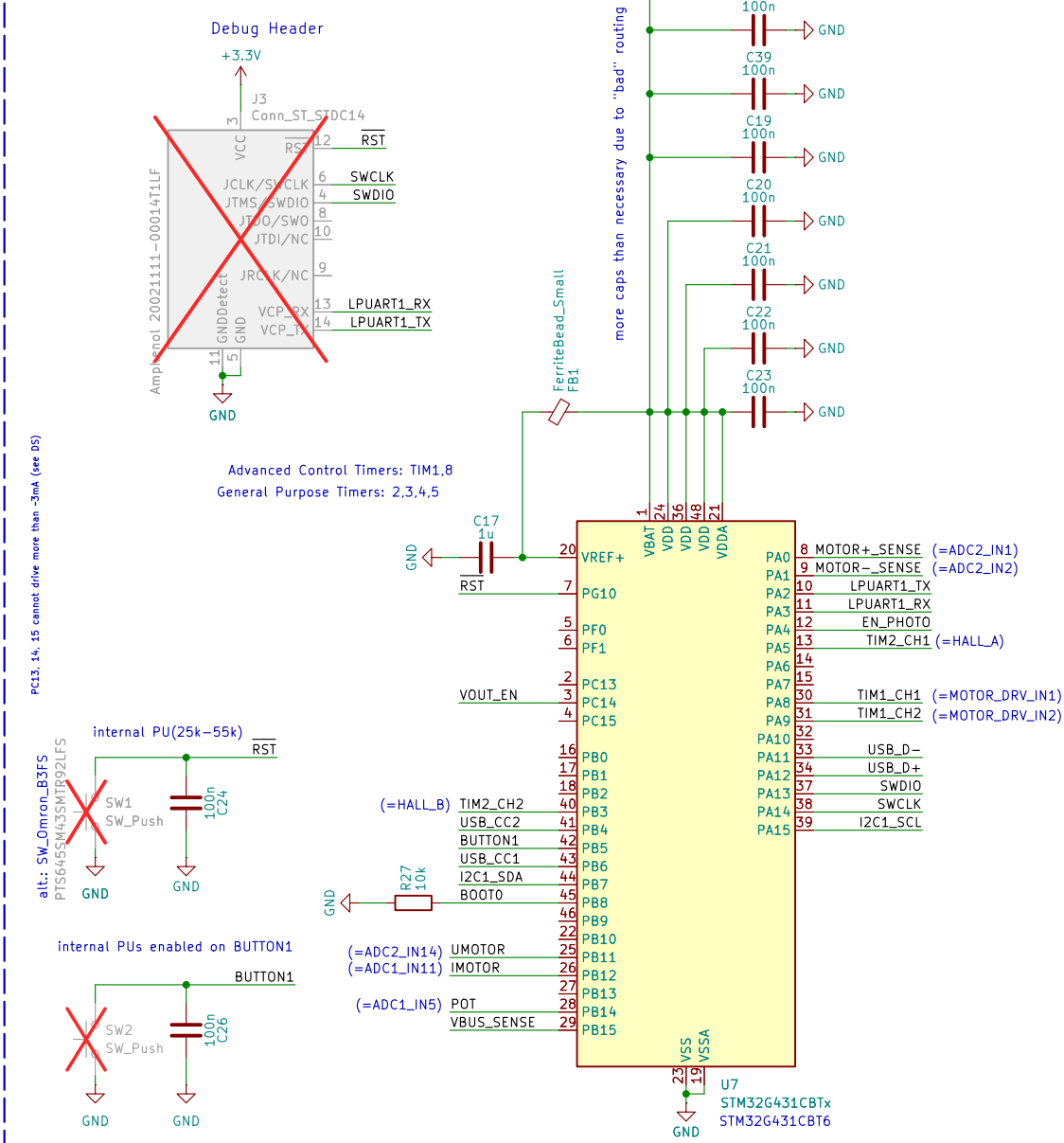
12V $\rightarrow$ 3.3V
-> potentially change to switcher LMR54406BVR
Inductor alternatives: ASP1-40205-100M-T (C3222037), SDCL1V4020-100M-R, 74405042100, XAL4040-103ME (C5144883)
handsolder: SPM4020T-100M-LR, ANR4020T100M (C6808085)



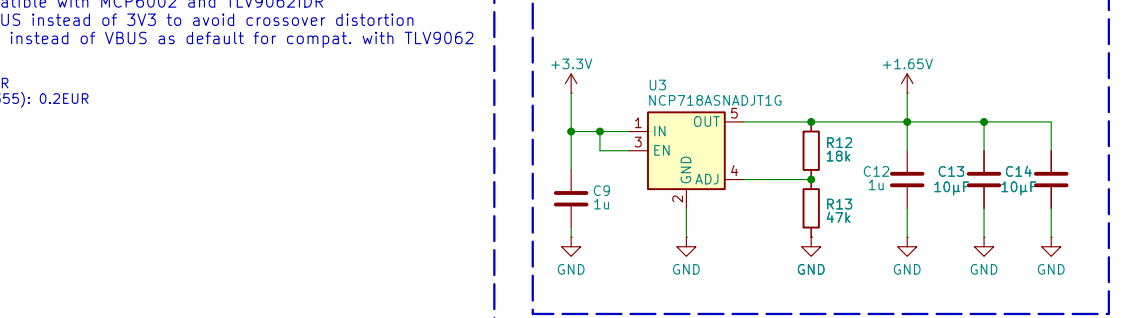
Back-EMF measurement



μC



Virtual GND for Back-EMF Measurement



5V for HALL-Supply and Back-EMF Measurement

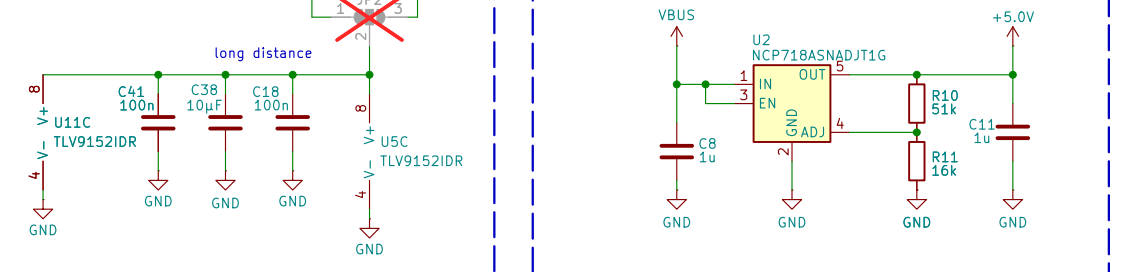
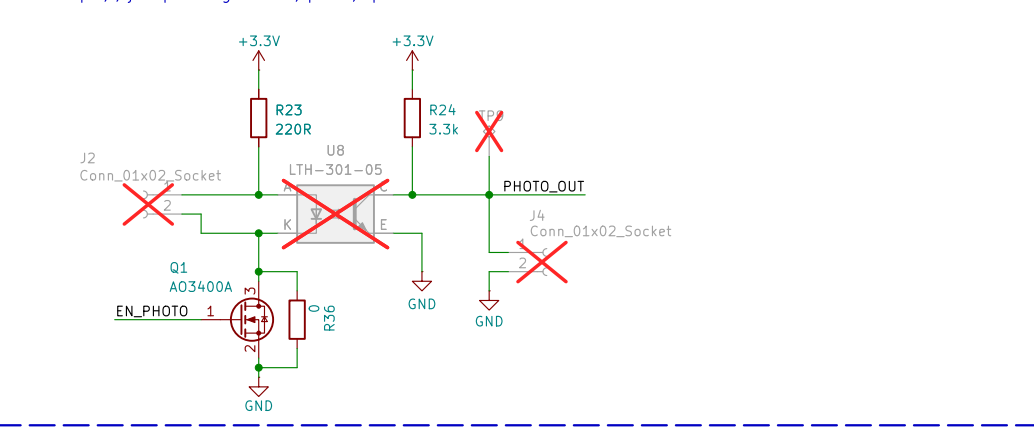


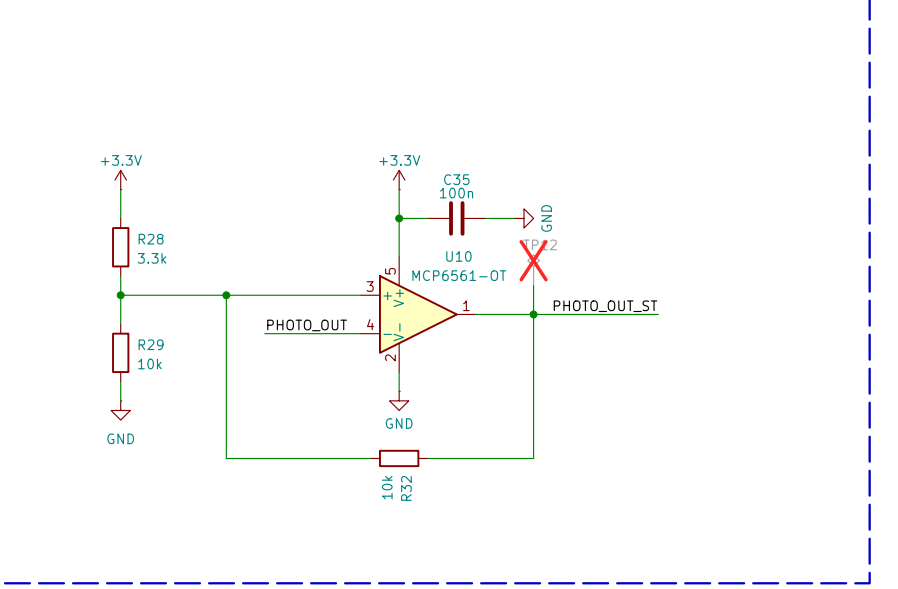
Photo Interruptor

No CTR given in Datasheet
1.)
At $I_F = 20mA \rightarrow$ Datasheet guarantees $> 0.5mA$ photocurrent $\rightarrow$ estimated worst case $CTR \sim 0.5/20 = 2.5\%$
 $\rightarrow$ worst case $IC @ 10mA$ LED drive $= 240\mu A$
 $\rightarrow$ For $R_{load} = 1k \rightarrow V_{low,out} = 3.3 - 240\mu A * 1k = 3.1V \rightarrow$ only 0.2V swing
2.)
To guarantee saturation in worst case, R_{load} must limit I_C to $240\mu A$
 $R_{load} > (3.3 - V_{CE,sat}) / 240\mu A = 13k\Omega$
3.)
Estimate R_{load} for large enough output swing:
For a swing of 2V $\rightarrow R_{load} = 2 / 240\mu A = 8.3k\Omega$
 $\rightarrow$ we want be rather fast, so try with a smaller R_{load} first
<https://josephehoff.github.io/posts/rpm>

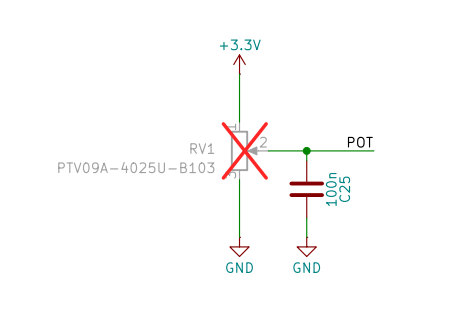


Schmitt-Trigger

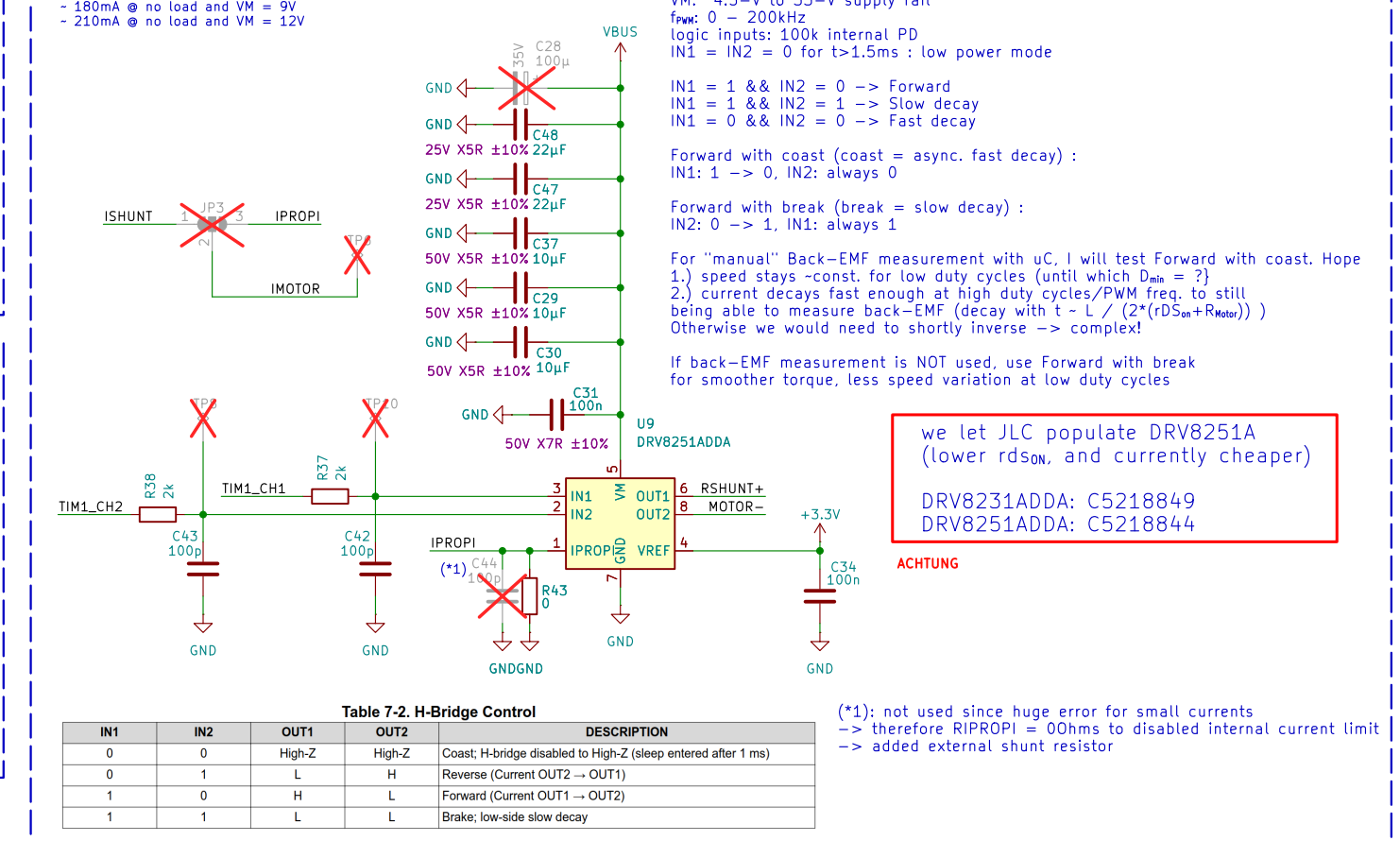
Inverting config
 $\rightarrow$ for Rthevenin $< R_{feedback}$: hysteresis-width - independent from threshold center
 $\rightarrow$ source impedance does not influence thresholds
with 1.5V swing from photointerruptor we expect 3.3V (high) $\rightarrow$ 1.8V (low)
 $\rightarrow$ set center threshold to $\sim 2.55V$
 $\rightarrow$ currently: switching threshold: 2.42V, hysteresis: 240mV



Potentiometer for manual speed control



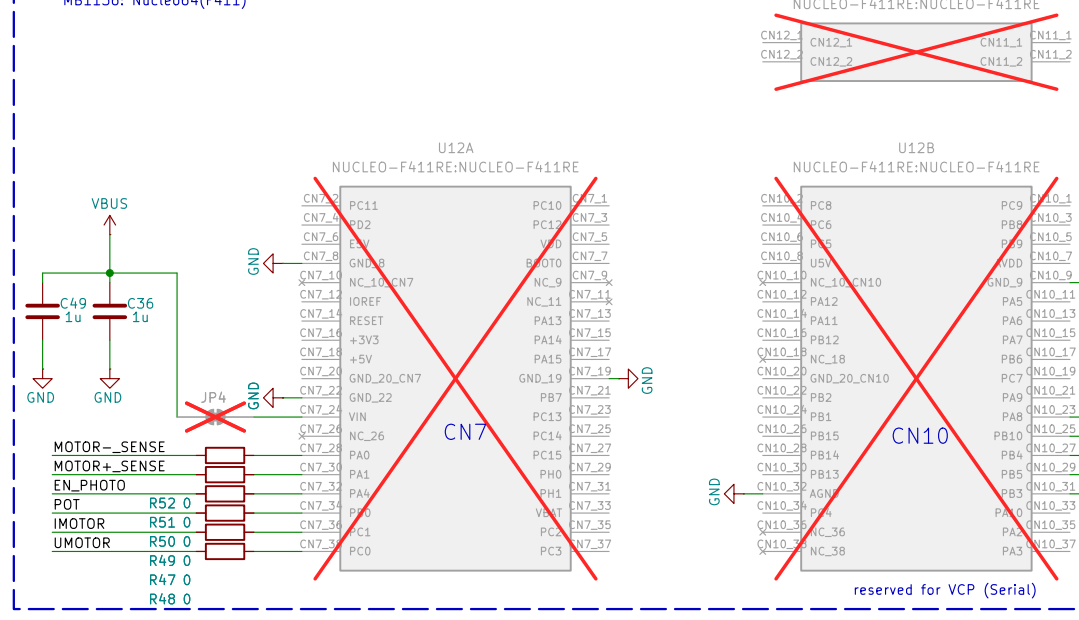
Motor-Driver



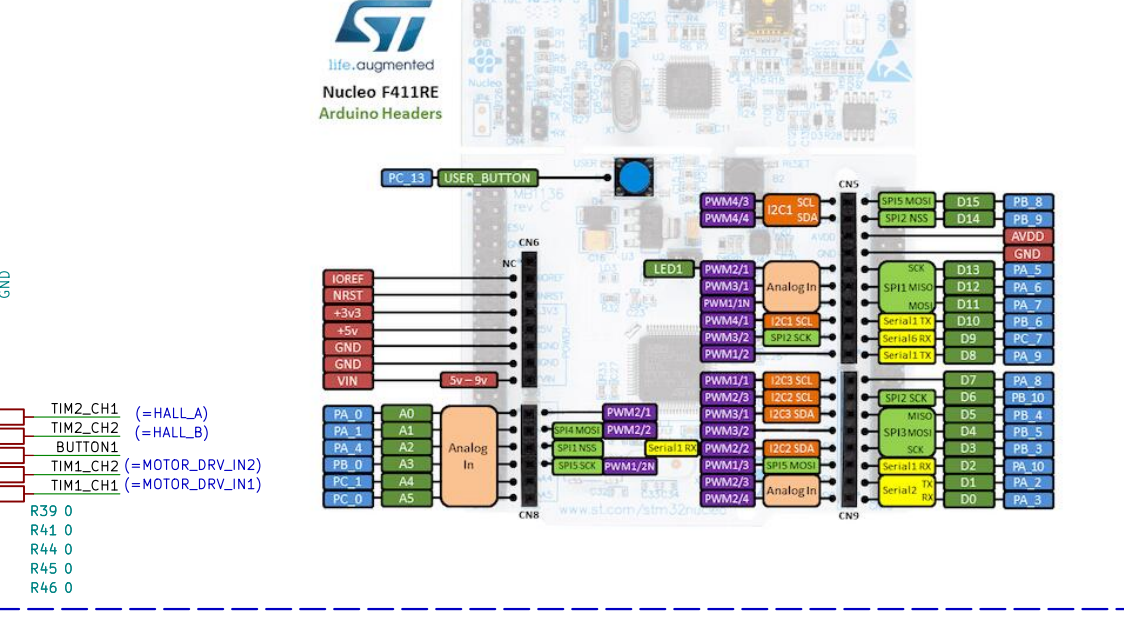
IN1	IN2	OUT1	OUT2	DESCRIPTION
0	0	High-Z	High-Z	Coast; H-bridge disabled to High-Z (sleep entered after 1 ms)
0	1	L	H	Reverse (Current OUT2 $\rightarrow$ OUT1)
1	0	H	L	Forward (Current OUT1 $\rightarrow$ OUT2)
1	1	L	L	Brake; low-side slow decay

(*) not used since huge error for small currents
 $\rightarrow$ therefore RIPROPI = 00ms to disabled internal current limit
 $\rightarrow$ added external shunt resistor

MATLAB

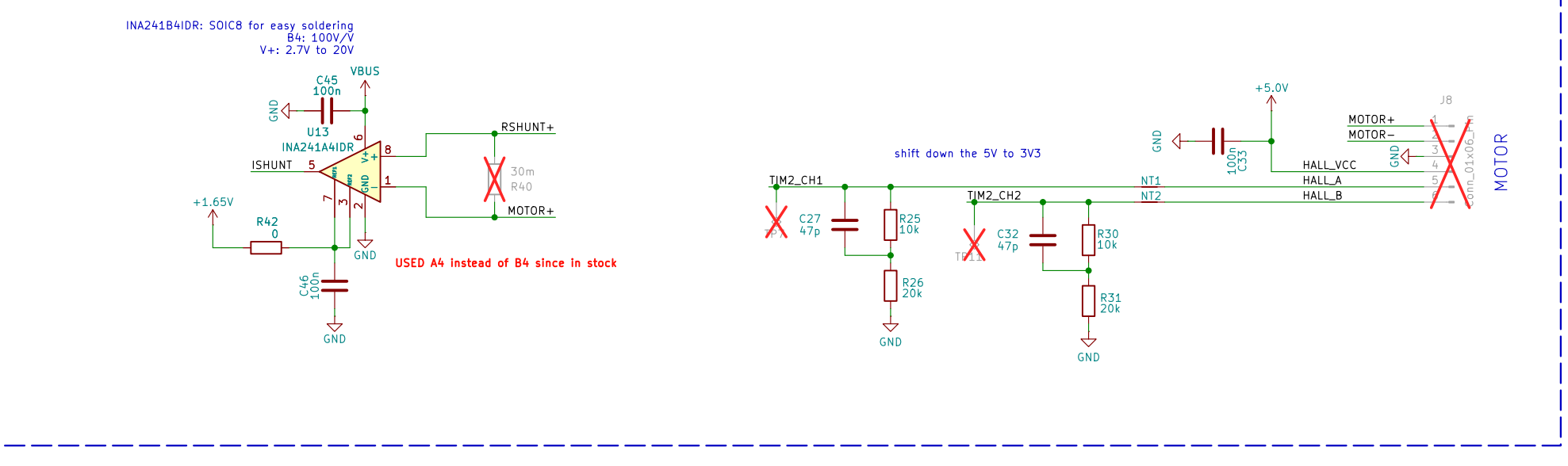


NUCLEO-F411RE:NUCLEO-F411RE



Motor Connector

pins mapped to allow TIM2 to be configured in CubeMX as "combined channels" in Encoder Mode to measure Motor Hall Encoder Signals as quadrature signals



Achtung:
Für $< 12V$: C15850 statt C440198
C1, C29, C30, C37 da für 12V ok und günstiger

DC Motor Driver Board
Elektronik & Technische Informatik
HTL Graz-Gösting
Autor: Franz Patz
Sheet: /
Title: DC Motor Driver Board
Size: A2 Date: 2024-11-30 Rev: V1.0
Kicad E.D.A. 10.0.6 Id: 1/1